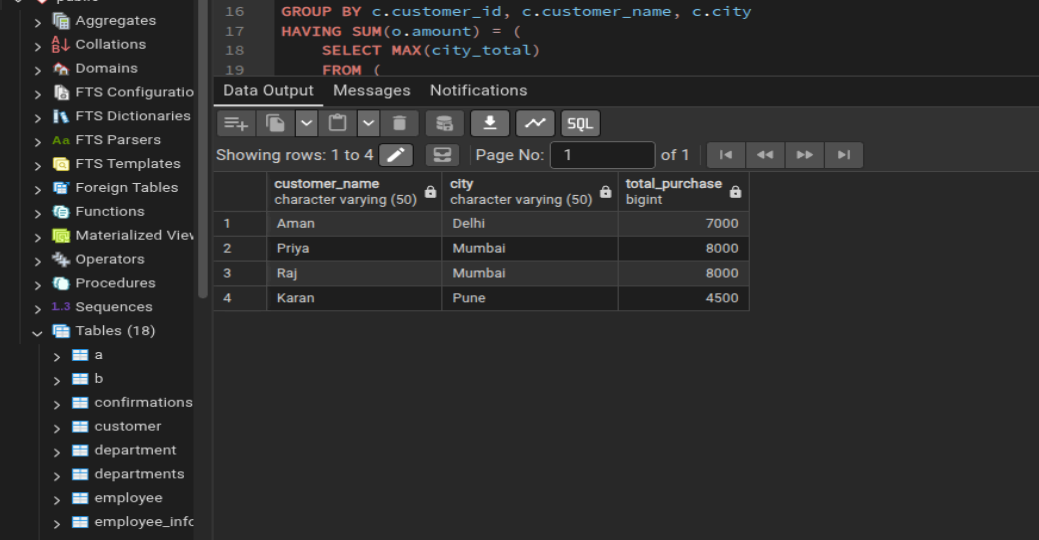


Lab mst dbms—

Question 1

```
SELECT c.customer_name,c.city,SUM(o.amount) AS total_purchase
FROM customer c
JOIN orders o ON c.customer_id = o.customer_id
GROUP BY c.customer_id, c.customer_name, c.city
HAVING SUM(o.amount)=(SELECT MAX(city_total)FROM (SELECT SUM(o2.amount) AS
city_totalFROM customer c2
JOIN orders o2 ON c2.customer_id = o2.customer_id
WHERE c2.city = c.city
GROUP BY c2.customer_id) AS city_totals
)ORDER BY c.city, c.customer_name;
```



The screenshot shows a database management tool interface. On the left is a sidebar with a tree view of database objects including Aggregates, Collations, Domains, FTS Configuration, FTS Dictionaries, FTS Parsers, FTS Templates, Foreign Tables, Functions, Materialized Views, Operators, Procedures, Sequences, and Tables (18). The main area displays an SQL query in a text editor. Below the editor is a 'Data Output' tab showing the results of the query. The results are presented in a table with four columns: customer_name, city, total_purchase, and an implicit row number. The data shows four rows of customer information and their total purchases.

	customer_name character varying (50)	city character varying (50)	total_purchase bigint
1	Aman	Delhi	7000
2	Priya	Mumbai	8000
3	Raj	Mumbai	8000
4	Karan	Pune	4500

Question 2

```
create or replace function product_add()
returns trigger as $$
begin
    insert into product_audit (product_id, product_name, action)
    values (new.product_id, new.product_name, 'rodct added');
    return new;
end;
$$ language plpgsql;
```

```
create trigger trg_product_added
after insert on product_inventory
for each row
execute function fn_product_added();
```

```
create or replace function fn_product_removed()
returns trigger as $$
begin
    insert into product_audit (product_id, product_name, action)
    values (old.product_id, old.product_name, 'remoed prodct');
    return old;
end;
$$ language plpgsql;
```

```
create trigger trg_product_removed
after delete on product_inventory
for each row
execute function fn_product_removed();
```

```
insert into product_inventory values (103, 'tablet', 25000);
delete from product_inventory where product_id = 101;

select * from product_audit;
```

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CREATE OR REPLACE FUNCTION product_add()

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RETURNS TRIGGER AS \$\$

Data Output

Messages

Notifications

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SQL

Showing rows: 1 to 2 Page No: 1 of 1

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	audit_id [PK] integer	product_id integer	product_name character varying (50)	action character varying (20)	action_time timestamp without time zone
1	1	103	Tablet	rodct added	2026-10-09 10:36:32.9266
2	2	101	Laptop	remoed prodct	2026-10-09 10:36:32.9266